

Fair-Neural Contextual Bandit based Recommendation System

Navigating the Fairness-Utility Trade-off in Recommendation Systems

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Abstract

Introduction Recommendation systems increasingly shape human experiences across digital platforms. These systems are also applicable in high-stakes domains such as medicine, employment, and economics. Current industry practices often prioritize performance over fairness, leading to amplifying demographic biases and marginalizing underrepresented content. The rapid integration of AI in society makes ethical considerations a critical priority, yet most existing research addresses either utility enhancement or bias mitigation in isolation. This study addresses this gap by integrating both objectives and demonstrating that high-performance neural architectures and rigorous fairness constraints can simultaneously achieve predictive accuracy and demographic parity. Specifically, this research introduces Fair-NeuralLinUCB, a novel contextual bandit model that enforces gender-based demographic parity without compromising the long-term stability of the recommendation system.

Research Question The primary research question is: “How does the integration of group-level exposure constraints into a neural contextual bandit framework empirically affect the trade-off between recommendation accuracy and algorithmic fairness?”

Method A Design Science Research approach was adopted with the two-stage Fair-NeuralLinUCB algorithm as the primary artifact. The model integrates an offline-pre-trained, frozen neural feature extractor with a fairness-constrained Linear Upper Confidence Bound (LinUCB) decision layer. Performance was evaluated through a model-based computational simulation on the MovieLens 1M dataset, using a pre-trained Singular Value Decomposition (SVD) Reward Oracle as the reward environment. The model was

benchmarked against unconstrained NeuralLinUCB and FairLinUCB models over the 50,000-step simulation, repeated across 10 random seeds to ensure statistical robustness.

Results The results show that Fair-NeuralLinUCB achieved the highest total reward ($37,107 \pm 477$) and the lowest cumulative regret ($11,454 \pm 487$). The model substantially reduced the gender-stereotype gap, and there was no statistically significant difference in utility relative to the NeuralLinUCB baseline. Furthermore, it significantly outperformed linear Fair-LinUCB in both reward and regret while achieving comparable fairness, indicating that neural architectures can effectively decouple the traditional fairness-accuracy trade-off.

Discussion These findings demonstrate that integrating fairness constraints into a neural contextual bandit does not statistically diminish predictive utility. This suggests that high-capacity feature extraction enables fairness penalties to operate as targeted adjustments rather than disruptive interventions. Consequently, this challenges the assumption of a “Price of Fairness,” indicating that ethically constrained neural recommendation systems are viable for deployment in demographically sensitive contexts without compromising performance. However, this study is limited by its reliance on a simulated reward environment and a single dataset. Future research focuses on investigating intersectional fairness, validating results in live deployment settings, and deriving formal theoretical regret bounds.

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Abbreviations

Abbreviation	Description
AI	Artificial Intelligence
NLP	Natural Language Processing
ML	Machine Learning
RS	Recommendation Systems
RL	Reinforcement Learning
CB	Contextual Bandits
CBF	Content-based Filtering
CF	Collaborative Filtering
SVD	Singular Value Decomposition
MDP	Markov Decision Process
UCB	Upper Confidence Bound
LinUCB	Linear Upper Confidence Bound
TS	Thompson Sampling
MLP	Multi-layer Perceptrons
DNN	Deep Neural Network
DSR	Design Science Research

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1

Introduction

In the age of artificial intelligence, recommendation systems play a pivotal role in shaping our daily experiences. Across diverse demographics, cultures, sectors, and businesses, these systems significantly influence individual and collective decision-making. Whether through product suggestions (e-markets), entertainment choices (such as movies, music, or digital content), social media feeds, news, politics, or even healthcare, we are continually guided by recommendation algorithms.

Despite their pervasive impact, the workings of these models often remain invisible to the general public. A critical research problem arises from this invisibility: most contemporary models are optimized solely for user engagement, which inadvertently leads to “bias amplification.” This often encodes and accelerates social injustices, creating “filter bubbles” and marginalizing niche or underrepresented content.

Hence, there is an urgent need to move beyond simple accuracy-based metrics and develop algorithmic frameworks that mathematically guarantee fairness without compromising recommendation utility. This study addresses this by integrating both objectives through high-performance neural architectures and rigorous fairness constraints. Specifically, this research introduces Fair-NeuralLinUCB, a novel contextual bandit model that enforces gender-based demographic parity to a neural recommendation system to address stereotype bias without compromising long-term stability.

A recommendation system is an information filtering tool designed to suggest items most relevant to a particular user. It gathers demographic information and data either implicitly (by tracking user behavior and interactions) or explicitly (through user ratings) [63]. Using advanced algorithms, it delivers personalized recommendations. Contextual bandits represent a new generation of recommendation systems, shifting from early static models to dynamic, online learning [68]. These models combine reinforcement learning principles with supervised learning, leveraging real-time contextual information to decide which item (arm) to recommend to a user. Contextual bandits are uniquely effective as recommendation systems because they treat recommendations as an iterative decision-making process rather than a static prediction task [8]. Bandits interactively explore new items to learn user preferences while simultaneously exploiting known information to provide relevant content. As the system receives user-click feedback, it continuously adapts its strategy to maximize a specific target (rewards). These models help mitigate the “cold start” problem of early static models [7] and are more efficient in producing output. The field has progressed significantly, evolving from the first contextual bandit model, Linear Upper Confidence Bound (LinUCB), used for news recommendation [35], to advanced neural UCB methods [67].

While the influence of recommendation systems on everyday life is immense, operating them without appropriate safeguards can raise serious concerns. These systems are trained on historical data, often sourced from user groups categorized by gender, race, age, income, and other demographic factors. Barocas et al. [6] argue that induction learning models, learning based on experience and data, are only as reliable as the data from which they are learned. Historical examples of such use cases will almost always reflect historical prejudices against certain social groups, dominant cultural stereotypes, and existing demographic inequalities. And using data with such identification patterns often means replicating the very same dynamics in automated systems. Recommendations can become biased and unfair, perpetuating existing inequalities. Tackling this unfairness of the system is key to building sustainable systems that add value and improve lives equally.

Fairness in bandit algorithms has been a longstanding area of research, with foundational work such as Joseph et al. [28] laying the groundwork.

An advancement relevant to this study is the FairLinUCB algorithm introduced by Huang et al. [24], which concentrates on promoting group fairness from the user perspective. The primary outcome of this model is its ability to ensure that the cumulative exposure of protected groups never falls below a predefined threshold. This demonstrates that it is mathematically possible to guarantee a minimum level of equity in a recommendation stream without significantly degrading the system’s predictive accuracy.

Incorporating fairness into recommendation systems introduces the concept of a fairness cost. For example, applying a fairness penalty to the LinUCB algorithm may prevent it from achieving the maximum possible reward. This trade-off between fairness and optimal reward is measured through ‘regret,’ which quantifies the loss in potential reward due to fairness constraints [24].

The NeuralLinUCB algorithm [53] is a recent advancement in contextual bandit methods for recommendation systems. It is a hybrid approach that integrates the nonlinear feature-extraction capabilities of deep learning with the principled exploration-exploitation decision-making of LinUCB methods. This blend allows recommendation systems to leverage both the representational power of neural networks and the reliability of linear models. However, neither structural complexity nor linearity inherently addresses algorithmic bias. Such a model will systematically replicate and amplify demographic inequalities present in historical data unless explicitly restricted [17]. With the development and practical application of FairLinUCB and NeuralLinUCB, it is now possible to combine these advancements by constructing a Fair-NeuralLinUCB model. This research aims to build and test such an algorithm.

1.1 Research Problem

Ideally, a recommendation system should be inherently unbiased and fair across demographic groups. However, machine learning algorithms often develop unintended biases that can perpetuate or amplify existing societal inequalities. While the field has progressed, a significant gap remains: existing fairness-aware models, such as FairLinUCB [24], are restricted by linear assumptions that fail to capture the complex, nonlinear relationships present in real-world user data. Conversely, state-of-the-art Neural models

[53] offer superior predictive accuracy but lack the necessary formal constraints to prevent the marginalization of minority groups. This study focuses specifically on neural contextual bandit-based recommendation systems.

The problem underlying this thesis is that modern contextual bandits designed to analyze complex features do not incorporate context-based fairness constraints. This thesis addresses this limitation by developing a Fair-NeuralLinUCB model. Within the scope of this research, algorithmic fairness is operationalized by focusing on mitigating group-based gender stereotype bias. By integrating deep neural feature extraction with a dynamic group-exposure constraint framework, this research provides a solution that bridges the gap between high-dimensional representation learning and measurable algorithmic equity.

1.2 Research Question

Previous research on fairness-aware contextual bandits and neural contextual bandits has largely remained siloed [11, 15]. To advance the field, it is essential to develop a hybrid fair neural contextual bandit that produces optimal outputs while maintaining algorithmic impartiality.

The primary research question of this study is: **How does the integration of group-level exposure constraints into a neural contextual bandit framework empirically affect the trade-off between recommendation accuracy and algorithmic fairness?**

To answer this, a comparative analysis is conducted by addressing the following two sub-questions:

- **Stereotype Mitigation:** To what extent does the Fair-NeuralLinUCB model reduce gender-based stereotype bias compared to an unconstrained NeuralLinUCB baseline?
- **Utility Preservation:** Can the Fair-NeuralLinUCB model maintain competitive recommendation performance in terms of cumulative regret and total reward compared to the linear FairLinUCB model, justifying the added complexity of the neural architecture?

2

Extended Background

In this section, we explore the related theories, key concepts, and past research findings on contextual bandit-based recommendation systems and fairness. Then, we define the theoretical framework for the Fair-NeuralLinUCB model by referring to past relevant studies.

2.1 Introduction to Research Background

The field of **Recommendation Systems (RS)** [7, 37, 51, 44, 47] serves as the economic and algorithmic engine for global platforms such as Netflix, YouTube, and Meta, where the precision of content delivery directly dictates user engagement, retention, and revenue. Historically, RS relied on static models trained on offline, historical datasets. However, these traditional approaches often fail to keep pace with the speed of modern digital environments, struggling particularly with the “cold start” problem and inefficiency. This led to a shift toward **Online Learning** systems, and with recent discoveries in **Reinforcement Learning (RL)** [55], **Contextual Bandits (CB)** [35, 13, 2] were established as a dominant framework for dynamic recommendation.

Despite their efficacy, RS models trained through iterative user feedback loops are susceptible to internalizing and amplifying societal biases present

in historical data. Since fairness in Artificial Intelligence (AI) emerges as a critical ethical and regulatory priority, the limitations of current RS frameworks have become increasingly evident. Existing **fairness-aware systems** [24, 6, 34] typically assume linear rewards. While this linearity ensures the mathematical tractability required for closed-form bias mitigation, these models lack the expressiveness necessary to navigate the high-dimensional, non-linear feature interactions inherent in modern complex user-item data.

Conversely, **Neural Bandit** [61, 68, 50] models leverage deep neural networks to achieve superior representational power and predictive accuracy. Yet, these architectures are fundamentally unconstrained with respect to fairness. Without an integrated mitigation mechanism, their sophisticated data interpretation can lead to systemic disparities. This thesis situates itself at the intersection of contextual bandits, recommender systems, and algorithmic fairness. It addresses a critical gap in the literature: the lack of fairness-aware neural contextual bandit algorithms that neutralize user-side unfairness without compromising recommendation utility.

2.2 Recommendation Systems

Recommendation systems emerged as a response to the internet’s information overload, evolving alongside the web itself. A typical RS comprises a database, a filtering algorithm, and a predictive model. Modern systems can deliver highly personalized content to users by leveraging both explicit data (such as direct user ratings) and implicit data (such as browsing or consumption behavior) [7]. Traditional recommendation systems commonly employ Content-based Filtering (CBF), Collaborative Filtering (CF), or a hybrid of both approaches.

Content-based Filtering [44, 47, 40] recommends new items based on the features of items the user has previously consumed, often resulting in suggestions like “Based on your watch history...”. However, this method can create “filter bubbles,” where users are repeatedly exposed to similar content. **Collaborative Filtering** [46, 51, 30] relies on user behavior data. If two users have interacted with the same item, the system assumes they share similar tastes and recommends future items accordingly, often with prompts like “Others who watched this also watched...”. This approach can struggle with the “cold start” problem, where insufficient data on new

users, items, or communities hampers the effectiveness of recommendations.

A major advance in CF is the matrix factorization technique, particularly Singular Value Decomposition (SVD) and its variants [30]. These methods decompose the sparse user-item rating matrix into lower-dimensional latent factor representations and predict ratings for unobserved user-item pairs as the dot product of their latent vectors. Although matrix factorization is insufficient as a standalone recommendation strategy because it cannot explore or adapt to new users, it serves a different, narrower function in this work: approximating user-item affinity scores from historical data to provide ground-truth probabilities for a stochastic binary reward environment in offline bandit simulation. MF serves as the basis for constructing the reward oracle detailed in Chapter 3.

Hybrid filtering methods were developed to address the above-mentioned limitations in CBF and CF-based systems [7, 9, 29, 1]. While hybrid approaches partially address the limitations of CBF and CF individually, they inherit a more fundamental problem — all three approaches treat recommendation as a static prediction task. The system is trained offline on historical interactions and deployed without adaptation. These systems often excel when users have substantial histories, there is considerable overlap in historical consumption among users, and the content universe is relatively static. This architectural assumption becomes untenable in dynamic environments where the digital landscape constantly changes, user preferences evolve, new items continuously enter the catalogue, and the system’s own recommendations shape future user behaviour through feedback loops [1]. Over time, users may lose interest if they are repeatedly shown similar recommendations, leading to decreased engagement. Another limitation is that positive feedback typically arrives only after many interactions. A system must make numerous recommendations before accumulating sufficient signal to reliably model user preferences [68, 35].

Hence, traditional recommendation systems face two significant challenges: managing the exploration-exploitation trade-off and adapting to the need for real-time, dynamic recommendations. Rather than optimizing for immediate, short-term gain, robust systems must prioritize long-term cumulative rewards [68]. In other words, a robust RS should not only exploit current knowledge of users but simultaneously explore to learn more about them [35]. It must strike a balance between recommending familiar items

and introducing new, diverse content. It should learn from user feedback and dynamically adapt its strategy to foster long-term engagement. This adaptability is especially critical for retaining new users with little or no historical data [12].

A **reinforcement learning system** addresses these challenges through a sequential decision-making framework known as the Markov Decision Process (MDP). In this process, an agent interacts with an environment over time; each action taken not only yields an immediate reward but also transitions the system to a new state, thereby influencing future opportunities and the cumulative return [68]. However, for many recommendation tasks, modeling the environment as a full MDP is often mathematically superfluous and computationally prohibitive, as the assumption that a single recommendation fundamentally alters the user’s permanent state introduces unmanageable complexity [35].

Additionally, a full RL system introduces significant computational overhead and algorithmic instability. These systems are frequently susceptible to the “deadly triad”—severe instability and divergence that occur when combining function approximation, off-policy learning, and bootstrapping [55, 64]. Consequently, complex RL models require resource-intensive components like experience replay buffers and continuous online backpropagation, making them notoriously difficult to stabilize in production environments [25].

2.3 Contextual Bandits Based Recommendation Systems

This thesis utilizes Contextual Bandits as a pragmatic middle ground. A CB system differs from full RL in that it is modeled as a one-step MDP, in which the agent does not consider the impact of the current action on the next state [20]. This “myopic” focus is highly suitable for recommendation systems where the primary goal is to map an immediate context vector to the best action without the need for complex state-transition learning [50]. Because they often allow for closed-form solutions or simple online updates—typically with $O(d^2)$ complexity—contextual bandits are more computationally efficient, stable, and easier to generalize than full RL counterparts [45]. Unlike traditional Recommendation Systems, which often rely

on static heuristics, CB models inherently address the exploration-exploitation dilemma, balancing the need to gather new information with the goal of maximizing immediate cumulative reward [3, 8].

In the CB framework, at each iteration, the agent observes an N -dimensional context (or feature vector) before selecting an action ('arm'). However, a defining limitation of this system is that the agent only receives one-sided feedback: the reward is observed strictly for the chosen arm, while the potential rewards for all other unselected arms remain unknown [33]. This partial observability creates a 'counterfactual' problem, where the system must estimate the value of actions it did not take [35]. Over time, the agent learns the underlying relationship between specific context vectors and their associated rewards, allowing it to accurately predict and select the optimal arm for any given context [2, 8].

2.3.1 Multi-Armed Bandit Algorithms

Before presenting CB algorithms, it is informative to review and compare foundational Multi-Armed Bandit exploration strategies, as LinUCB (the algorithm used in this study) is best understood as a contextual extension of the UCB framework.

Traditional Multi-Armed Bandit algorithms are often context-free, meaning they do not leverage side information (features) to inform their decisions. Common foundational algorithms in this category include **ϵ -greedy**: A simple yet effective heuristic that selects a random arm with probability ϵ and the best-performing arm otherwise [32]. **Upper Confidence Bound (UCB)**: A deterministic approach that accounts for uncertainty by selecting arms with the highest potential reward upper bound [3]. **Thompson Sampling (TS)**: A Bayesian approach that selects arms by sampling from a probability distribution maintained for each arm's expected reward [56, 2, 49].

Although ϵ -greedy is simple to implement, its uniform exploration approach (blind exploration) without regard to uncertainty wastes the exploration budget on arms that are already well understood. Keeping ϵ constant leads the agent to continue making suboptimal choices rather than identifying the optimal policy, ultimately reducing total reward. Choosing the "right"

ϵ value is difficult; too high results in excessive exploration and slow learning, while too low leads to premature convergence and under-exploration [32, 55]. Thompson Sampling often outperforms deterministic algorithms such as UCB, but its reliance on probability distributions and random sampling incurs significant computational overhead and heightens sensitivity to modeling assumptions, especially in high-dimensional feature spaces. Moreover, it does not straightforwardly extend to the neural representation setting on which this study is based [49].

The UCB framework is built on the principle of “optimism in the face of uncertainty.” It assigns a score to each arm based on two components:

The Estimated Mean ($\hat{\mu}$): The average reward the arm has yielded in the past, representing exploitation.

The Confidence Interval (Bound): A mathematical measure of uncertainty representing exploration.

The algorithm consistently selects the arm with the highest total score. As an arm is pulled more frequently, its confidence interval shrinks—reflecting increased certainty—eventually leading the algorithm to favor arms with high proven rewards.

Standard UCB (or “Vanilla” UCB) treats every arm as a completely independent entity. It aims to find the “globally best” arm, essentially determining which content is most popular on average for the entire user base, rather than the best arm for the current user. Its complexity is strictly tied to the number of arms K , and it lacks the capacity to differentiate between users. Therefore, Vanilla UCB, while foundational, is unsuitable for personalized recommendations because its arm scores are context-free.

To overcome this, the LinUCB algorithm, which incorporates a feature vector $x_{t,a}$, was developed by [Li et al. \(2010\)](#) as a contextual extension of the UCB framework. This provides three primary advantages. **Personalization:** By including user-specific data (age, history, location) in the context vector, LinUCB learns that the “best arm” for User A may differ from User B, even within the same set of available arms. **Addressing the Cold-Start Problem:** LinUCB can estimate the potential reward for new items based on their metadata features, whereas Vanilla UCB must pull a new arm multiple times before it can make any informed decision [57]. **Information Generalization:** In systems with similar items, LinUCB reaches

the optimal policy faster (lower cumulative regret) because learning about one arm’s features informs the model about other arms with similar features. Most importantly for this thesis, LinUCB’s additive score structure — exploitation term plus exploration bonus — provides a natural insertion point for the fairness correction. The fairness bonus can be added directly to the UCB score without disrupting the algorithm’s theoretical regret guarantees, an integration that is not achievable in sampling-based selection of Thompson Sampling or ϵ -greedy’s random exploration.

The LinUCB algorithm operates under a key assumption: that the relationship between features and rewards is linear (hence the name LinUCB). If user preferences are highly complex or non-linear, this may result in a higher approximation error than with more expressive models [13]. This study partially mitigates this limitation by the neural feature extractor $\phi(x)$, which projects raw user-item interactions into a non-linear representation space before the linear bandit operates on it. The neural feature extraction pipeline is detailed in **Section 2.4**.

2.3.2 Contextual Multi-Armed Bandit process

Following the framework established by Li et al. [35], the CB process operates in discrete trials $t = 1, 2, 3, \dots, T$. In each trial t , the algorithm executes the following sequence.

Observation: The algorithm observes the current user u_t and a set of available arms A_t (actions). For each arm $a \in A_t$, a feature vector $x_{t,a}$ is provided. This vector, known as the context, summarizes relevant information about both the user and the specific arm.

Selection & Feedback: Based on previous observations and historical payoffs, the algorithm selects an arm $a_t \in A_t$. It then receives a payoff (reward) r_{t,a_t} , the expectation of which is determined by the relationship between the context x_{t,a_t} and the chosen action a_t .

Optimization: The algorithm updates its arm-selection strategy using the new observation $(x_{t,a_t}, a_t, r_{t,a_t})$. Crucially, the algorithm only observes the payoff for the chosen arm; the payoffs for all other unselected arms $a \neq a_t$ remain unknown.

The primary objective of the algorithm is to maximize the expected total payoffs over T trials, defined as $\sum_{t=1}^T r_{t,a_t}$. In the bandit literature, this is more commonly expressed as minimizing cumulative regret [35].

Let a_t^* be the optimal arm that yields the maximum expected payoff at trial t . The optimal expected T -trial payoff is defined as:

$$E \left[\sum_{t=1}^T r_{t,a_t^*} \right]$$

The T -trial regret $R_A(T)$ of algorithm A is formally defined as the difference between the expected payoff of the optimal strategy and the payoff actually achieved by the algorithm:

$$R_A(T) = E \left[\sum_{t=1}^T r_{t,a_t^*} \right] - E \left[\sum_{t=1}^T r_{t,a_t} \right]$$

By minimizing this regret, the algorithm learns to balance the exploration of new arms with the exploitation of known high-reward arms, eventually converging toward the optimal policy. In standard offline settings, however, a_t^* is unobservable; only the reward for the selected arm is logged, making regret computation directly from data impossible. In this thesis, the optimal arm a_t^* , at each timestep, is defined as the arm that yields the highest predicted reward among the candidate set, as determined by the SVD reward oracle. This definition is tractable because matrix factorisation produces dense predictions across all user-item pairs, including unobserved interactions.

LinUCB is typically implemented in one of two architectures. **Disjoint Linear Model:** Each arm is treated independently with its own set of parameters. **Hybrid Linear Model:** Features are split into those specific to an arm and those shared across all arms. This allows for information sharing, which is particularly effective in sparse data scenarios where some items have very few interactions.

LinUCB for Disjoint Linear Models

This thesis implements the disjoint LinUCB model described by Li et al. [35], as detailed in Algorithm 1, where each arm maintains independent

parameters. This choice is appropriate because the neural feature extractor $\phi(x_{t,a})$ already captures shared user-item interactions in the representation space, making the hybrid model's shared coefficient redundant. The bandit, therefore, only needs to learn arm-specific reward mappings on top of that representation.

Algorithm 1: LinUCB with disjoint linear models (adapted from Li et al. (2010)[35]).

Inputs: $\alpha \in \mathbb{R}_+$

```

1: for  $t = 1, 2, 3, \dots, T$  do
2:   Observe features of all arms  $a \in \mathcal{A}_t : \mathbf{x}_{t,a} \in \mathbb{R}^d$ ;
3:   foreach  $a \in \mathcal{A}_t$  do
4:     if  $a$  is new then
5:        $\mathbf{A}_a \leftarrow \mathbf{I}_d$  ( $d$ -dimensional identity matrix);
6:        $\mathbf{b}_a \leftarrow \mathbf{0}_{d \times 1}$  ( $d$ -dimensional zero vector);
7:        $\hat{\boldsymbol{\theta}}_a \leftarrow \mathbf{A}_a^{-1} \mathbf{b}_a$ ;
8:        $p_{t,a} \leftarrow \hat{\boldsymbol{\theta}}_a^\top \mathbf{x}_{t,a} + \alpha \sqrt{\mathbf{x}_{t,a}^\top \mathbf{A}_a^{-1} \mathbf{x}_{t,a}}$ ;
9:   Choose arm  $a_t = \arg \max_{a \in \mathcal{A}_t} p_{t,a}$  with ties broken arbitrarily,
   and observe a real-valued payoff  $r_t$ ;
10:   $\mathbf{A}_{a_t} \leftarrow \mathbf{A}_{a_t} + \mathbf{x}_{t,a_t} \mathbf{x}_{t,a_t}^\top$ ;
11:   $\mathbf{b}_{a_t} \leftarrow \mathbf{b}_{a_t} + r_t \mathbf{x}_{t,a_t}$ ;
```

The expected payoff of an arm a at trial t is modeled as:

$$E[r_{t,a} | x_{t,a}] = x_{t,a}^\top \theta_a^*$$

The disjoint model maintains independent parameters for each arm, meaning θ_a^* is estimated solely from interactions involving arm a , and this prevents information sharing across arms. The core building blocks of the algorithm, which are the arm-specific covariance matrices ($\mathbf{A}_a$) and the response vectors ($\mathbf{b}_a$), have fixed dimensions based on the feature space. These components can be updated incrementally using simple additive operations at each trial t . This eliminates the need to store the entire history of interactions in an expensive experience replay buffer, a requirement that often plagues Full RL systems.

The algorithm is highly scalable because the computations at any given time t are restricted only to the arms available in the current candidate

set $\mathcal{A}_t$. Quantities associated with arms not present in the current context are excluded from the calculation, enabling the system to handle large, dynamic item catalogs efficiently. While calculating the UBC involves matrix inversion, the per-trial computational complexity can be significantly reduced. The per-trial complexity is maintained at $O(d^2)$, where d is the dimension of the arm-specific feature spaces.

2.4 Neural Bandits

As discussed in Section 2.3, the fundamental linearity assumption underlying LinUCB substantially limits its ability to model complex user-item interactions. In real-world recommendation scenarios, the context for determining the optimal action consists of heterogeneous information about the user, the item, and the latent dynamics between them. This feature space is inherently nonlinear. User preferences and item characteristics cannot be captured by simple linear transformations, as users exhibit complex, evolving demographic profiles and items are characterized by high-dimensional feature sets.

To overcome these limitations, deep learning-based recommendation systems have been introduced. Deep Neural Networks (DNNs) effectively capture non-linear and non-trivial relationships and excel at identifying intricate patterns across diverse data streams by leveraging extensive contextual, textual, and visual information [65]. The incorporation of deep learning into recommendation systems provides several advantages, including nonlinear data transformation, automated representation learning, sequence modeling, and architectural flexibility.

However, these strengths are accompanied by a significant trade-off: deep networks frequently operate as “black-box” models. This lack of transparency creates considerable challenges for explainability and interpretability [65]. This thesis does not resolve this limitation — the neural feature extractor $\phi(\mathbf{x}; \mathbf{w})$ remains opaque in terms of which input features drive its representations.

2.4.1 Neural Recommendation Systems

Neural network-based recommendation systems have been extensively implemented in traditional architectures (e.g., collaborative filtering) as well as in RL-based models [63]. In this thesis, we focus on leveraging a neural network for high-dimensional feature extraction, which is subsequently transferred to a contextual bandit framework, specifically, the LinUCB algorithm.

Multi-layer Perceptrons (MLP)

The Multi-layer Perceptron (MLP) is a common type of DNN that serves as the backbone of our neural contextual bandit. As a feed-forward neural network with one or more hidden layers, the MLP employs non-linear activation functions to model data as a stack of transformations, learning hierarchical feature representations. Due to their universal approximator property, MLPs are uniquely capable of capturing the complex, non-trivial relationships inherent in user-item contexts [65].

Evolution of Neural Bandit Models

Neural bandit models have been widely studied across various algorithms, including Thompson Sampling and LinUCB [67, 66, 4, 5]. Riquelme et al. [49] established the foundational principle of an architecture in which the neural network is trained for representation, with a linear method applied on top. By approximating Bayesian neural networks within a Thompson Sampling framework, they demonstrated that decoupling representation learning from decision-making provides a robust balance between exploration and exploitation.

The adoption of LinUCB into a similar setting is introduced by Xu et al. [60]. In this framework, the neural network performs a nonlinear feature mapping, $\phi(\mathbf{x}; \mathbf{w})$, which serves as input to a LinUCB head. The process is formalized as:

$$f(\mathbf{x}) = \boldsymbol{\theta}^{*\top} \phi(\mathbf{x}; \mathbf{w})$$

θ^* is the optimal linear weight vector learned by the LinUCB head, and $\top$ denotes the transpose operation. The predicted reward $f(\mathbf{x})$ is therefore a linear function of the neural representation, combining non-linear feature extraction with a tractable linear decision layer. This formulation decouples the representation from the exploration weights. The LinUCB head computes a UCB score comprising an exploitation term (predicted reward) and an uncertainty bonus derived from the last-layer representation. To maintain efficiency in online settings, the neural network backbone is re-trained periodically (e.g., after a set number of epochs), while the linear head updates its covariance matrix A_t —which accumulates the outer products of the representations—continuously.

This “Neural-Linear” method is often more computationally tractable for large-scale recommendation systems, as it avoids the memory overhead of maintaining high-dimensional gradient-based covariance matrices. Despite the success of this model, adaptive retraining of the neural backbone introduces a significant technical hurdle known as catastrophic forgetting [62]. When a model is retrained on new data, the shifting latent representations $\phi(\mathbf{x}; \mathbf{w})$ effectively invalidate the accumulated A_a statistics in the bandit head. Since the precision matrix and weight vectors are geometrically tied to the specific feature space at the time of observation, any change to the underlying neural weights creates a mismatch between historical statistics and current features. While Zahavy and Mannor [62] suggested mitigating this by using a limited-memory version that matches the likelihood of old features, this approach adds significant computational complexity to online systems.

2.4.2 The Two-Stage DeepLinUCB Architecture

To circumvent these stability issues, the architecture adopted in this thesis most closely follows the DeepLinUCB framework of Shi et al. [53]. They propose a two-stage design: training a deep neural network offline on historical interaction data using binary cross-entropy loss as shown in Algorithm 2, freezing its weights, and feeding the last-layer representation directly into a disjoint LinUCB head for online serving as explained in Algorithm 3.

Algorithm 2: DeepLinUCB Offline part (adapted from Shi et al.(2023)[53])

Input : Number of batches U , samples in each batch n , learning rate η , contexts $\{\mathbf{x}_{i,a_i}\}_{i=1}^{U \cdot n}$, rewards $\{r_{i,a_i}\}_{i=1}^{U \cdot n}$, initial parameter $\phi^{(0)}$.

- 1: CE denotes the cross entropy loss;
 - 2: **for** $j = 0, \dots, J - 1$ **do**
 - 3: $\mathcal{L}_j(\phi) = \sum_{t=j \cdot n + 1}^{(j+1) \cdot n} CE(f(\mathbf{x}'_{t,a_t}; \phi), r_{t,a_t});$
 - 4: $\phi^{(j+1)} = \phi^{(j)} - \eta \nabla \mathcal{L}_j(\phi^{(j)});$
 - 5: **return** $f_{\phi^{(J)}}$;
-

Algorithm 3: DeepLinUCB Online part (adapted from Shi et al.(2023)[53])

Inputs: Trained DNN model $f_{\phi'}$'s last layer output as $f_{\phi'}$, exploration parameter $\alpha \in \mathbb{R}_+$.
Denote the length of last layer of DNN f_{ϕ} as d .

- 1: **for** $t = 1, 2, 3, \dots, T$ **do**
 - 2: Observe features of all arms $a \in \mathcal{A}_t : \mathbf{x}'_{t,a};$
 - 3: Calculate the input of LinUCB via the frozen DNN
 $\mathbf{x}_{t,a} = f_{\phi'}(\mathbf{x}'_{t,a}) \in \mathbb{R}^d;$
 - 4: **for all** $a \in \mathcal{A}_t$ **do**
 - 5: **if** a *is new* **then**
 - 6: $\mathbf{A}_a \leftarrow \mathbf{I}_d$ (d -dimensional identity matrix);
 - 7: $\mathbf{b}_a \leftarrow \mathbf{0}_{d \times 1}$ (d -dimensional zero vector);
 - 8: $\hat{\boldsymbol{\theta}}_a \leftarrow \mathbf{A}_a^{-1} \mathbf{b}_a;$
 - 9: $p_{t,a} \leftarrow \hat{\boldsymbol{\theta}}_a^\top \mathbf{x}_{t,a} + \alpha \sqrt{\mathbf{x}_{t,a}^\top \mathbf{A}_a^{-1} \mathbf{x}_{t,a}};$
 - 10: Choose arm $a_t = \arg \max_{a \in \mathcal{A}_t} p_{t,a}$, observe a real-valued payoff
 $r_t;$
 - 11: $\mathbf{A}_{a_t} \leftarrow \mathbf{A}_{a_t} + \mathbf{x}_{t,a_t} \mathbf{x}_{t,a_t}^\top;$
 - 12: $\mathbf{b}_{a_t} \leftarrow \mathbf{b}_{a_t} + r_t \mathbf{x}_{t,a_t};$
-

In this setup, uncertainty estimates are updated only for the arm selected at each timestep. This design—offline supervised pre-training followed by frozen feature extraction—is validated at production scale in the Hulu video recommendation system [53], where it demonstrably outperforms both standalone LinUCB and standard DNN baselines.

Consistent with this, the thesis adopts the disjoint LinUCB formulation of Li et al. [35], maintaining independent per-arm covariance matrices A_a rather than a single shared matrix A_t . This framework is adapted for an offline simulation setting as follows.

Stage 1: Representation Learning: The network is pre-trained as a binary classifier on oracle-labeled user-movie pairs. By deriving labels directly from an SVD oracle rather than raw rating thresholds, the representation learning objective is closely aligned with the bandit’s reward signal. **Stage 2: Frozen Execution:** Once trained, the weights are frozen and rendered non-trainable, and the model is set to evaluation mode to disable dropout and fix Batch Normalization statistics. This ensures that $\phi(\mathbf{x}; \mathbf{w})$ serves as a static feature extractor throughout the simulation, preserving the integrity of the A_a matrices and preventing the representation drift [62]. Furthermore, the additive UCB score structure — exploitation term plus exploration bonus — is preserved in this frozen architecture, providing the clean insertion point for the fairness correction developed in Chapter 3.

2.5 Fairness

Fairness is a multifaceted concept that lacks a single, universal definition. Consequently, a truly fair system must account for context-dependent interpretations. Despite this complexity, non-discrimination and equality serve as the foundational pillars of fairness in algorithmic systems. According to the Alan Turing Institute’s guidelines on ethical AI, these concepts are predicated on the core belief in the equal moral status of all human beings. This entails a right to equal respect, protection, and regard before the law—principles that must be mathematically and procedurally encoded into AI models to prevent the amplification of historical biases [34].

A significant unintended consequence of automated system design is the emergence of systemic unfairness [6]. While machine learning systems may appear neutral—given their foundation in mathematical logic—they are ultimately products of human design and oversight. Consequently, these

systems are susceptible to the inherent limitations of their creators’ contexts, as well as the cognitive and social biases that influence human interaction. Pre-existing historical discrimination and structural social injustices—and the prejudices they foster—can be inadvertently integrated into the AI innovation lifecycle. These biases can manifest at any stage of the project workflow [54].

Fairness in AI systems is a multifaceted challenge that can be addressed at various stages of the development pipeline. Following the framework established in the Survey on Fairness-aware Recommender Systems [26], these interventions are systematically categorized into three primary phases. **Pre-processing:** This stage focuses on mitigating bias within the training data before it reaches the model. Common techniques include data re-labeling, resampling, or weighting to ensure that historical imbalances or under-representations are corrected at the source. **In-processing:** Fairness objectives are integrated directly into the model’s learning phase. This is achieved through methods such as regularization (penalizing biased outcomes in the loss function), causal inference, and adversarial debiasing, which force the algorithm to learn patterns that are independent of protected or sensitive attributes. **Post-processing:** These techniques are applied to the model’s output after the initial predictions have been generated. In recommendation contexts, this typically involves re-ranking algorithms or calibrated recommendations designed to ensure the final list of suggestions meets specific diversity or parity constraints. By systematically addressing fairness at each stage, developers can move beyond simple mathematical logic to create AI systems that are both more equitable and more reliable in real-world applications.

To address fairness comprehensively, it is essential to categorize biases by the development stage at which they emerge. In the pre-processing stage, the focus is on **dataset bias**, which includes user attribute and selection bias—where demographic data is skewed or non-representative—and **exposure bias**, which occurs when users interact only with a limited subset of items. This stage also contends with **time and cold start bias**, which creates discrepancies in how established versus new items or users are treated. As the system moves to the in-processing stage, **model bias** becomes the primary concern, specifically ranking bias, in which the algorithm inherently favors certain positions or attributes during scoring. Finally, the post-processing stage addresses **feedback bias**, most notably **popularity bias** (often referred to as the Matthew Effect). This “rich-get-richer” phenomenon

ensures that already popular items receive disproportionate exposure, further marginalizing niche content and reinforcing historical inequities in a continuous loop [26, 42, 36].

2.5.1 Fairness in Recommendation Systems

Fairness in RS differs significantly from other Machine Learning applications due to the interplay between multiple stakeholders and the dynamic nature of the feedback loop. **Multi-Sided Fairness:** An RS must account for both users and the items (or providers) they consume. **User-Side (Consumer) Fairness** ensures that distinct individuals or demographic groups receive an equivalent quality of service. Conversely, **Item-Side (Provider) Fairness** focuses on product exposure (or “arms”), aiming to mitigate popularity bias and ensure equitable visibility for all providers. **Dynamic Evolution:** Unfairness can manifest and evolve throughout the entire lifecycle of an RS due to continuous interactions. Without proactive fairness regulations at each stage of development, the system risks learning historical inequities from previous cycles and amplifying them in subsequent feedback loops. An RS may exhibit bias in its decision-making through either intended or unintended mechanisms. Intended bias occurs when a system is deliberately trained to be selective, for example, by prioritizing paid promotions. Unintended bias arises from algorithmic flaws or historical data that reflects societal prejudices. As reliance on these systems grows, their potential for negative societal impact increases [26, 36]. This thesis focuses on single-sided user fairness, specifically whether users of different genders receive equitable movie recommendations in their preferred genres.

Fairness in RS is categorised by both the scope and the timeframe of the intervention. At the metric level, the primary distinction lies between Group Fairness and Individual Fairness. **Group fairness** requires that a system’s outcomes are equitably distributed across demographically defined groups. The most widely adopted group fairness criterion is **demographic parity**, which requires that the probability of a positive outcome is independent of group membership: $P(\hat{Y} = 1 \mid A = 0) = P(\hat{Y} = 1 \mid A = 1)$, where A denotes protected attribute membership such as gender or age group [42, 6]. In recommendation systems, where items are ranked and selected rather than classified, this principle translates to equal exposure — each demographic group G_i should receive recommendations at a rate no lower than

a predefined threshold L_i . This thesis operationalises demographic parity at the genre level rather than at the overall recommendation level. For each genre g and each gender group, the fairness tracker monitors whether the recommendation rate gap exceeds a threshold ϵ , applying corrective bonuses to underserved gender-genre combinations.

A stricter criterion, **equal opportunity** [21], requires that true positive rates are equalized across groups — that is, qualified candidates from all groups have an equal probability of receiving a positive recommendation. While demographic parity constrains overall exposure rates, equal opportunity additionally requires that the quality of recommendations is consistent across groups, penalizing systems that meet exposure targets by recommending low-reward items to minority groups.

Individual fairness, formalised by Dwork et al. [16], adopts a complementary perspective: similar individuals should be treated similarly, regardless of group membership. Formally, if two users u and v are similar under some task-specific metric $d(u, v)$, the system’s recommendations for them should also be similar. Individual fairness is often operationalised through counterfactual fairness — a recommendation is counterfactually fair if it would remain unchanged had the user belonged to a different demographic group, holding all other attributes constant [31]. While individual fairness provides stronger per-user guarantees than group-level constraints, it is harder to enforce in practice as it requires a meaningful similarity metric over the user space. Focusing on demographic parity alone can inadvertently impose a “fairness tax” at the individual level; by forcing group-level equilibrium, the system may reinforce demographic stereotypes rather than catering to the user’s idiosyncratic preferences. This is identified as a limitation of this study.

Systems are further evaluated on whether their approach is static or dynamic. Static fairness aims to provide immediate, equitable outcomes regardless of environmental changes. Dynamic fairness accounts for temporal factors to maintain equity over time — a distinction particularly relevant in bandit settings where exposure accumulates across timesteps and short-term constraint violations may compound into long-term inequity [36, 26]. This thesis adopts a dynamic fairness perspective, monitoring cumulative group exposure $N_i(T)$ across the simulation horizon rather than enforcing per-timestep constraints.

2.5.2 Fair Contextual Bandits

Within the Full RL paradigm, several sophisticated frameworks have been introduced to balance utility and equity, such as FairRec [38], Counterfactual Explainable Fairness and Pareto Efficient Fairness-Utility [19], and the multi-objective MDP-based framework Popcorn [18]. However, these systems often carry the computational weight and instability inherent in Full RL, making them impractical for recommendation settings that require lightweight, interpretable, and incrementally updatable fairness mechanisms.

This thesis, therefore, situates its fairness intervention within the contextual bandit framework, which offers a more tractable and theoretically grounded alternative. CB algorithms are uniquely positioned to mitigate bias due to their online learning nature, which allows for the correction of biases as they emerge in real-time. By maintaining a balance between exploration and exploitation, CB models prevent the system from collapsing into a deterministic ‘popularity trap.’ However, while the inherent exploration mechanism provides a foundation for fairness, CB operates on individual-arm uncertainty rather than group-level exposure and cannot guarantee that demographic groups meet their target exposure rates [24]. Explicit interventions are required to ensure that the selection process adheres to specific equity constraints rather than purely maximizing cumulative reward.

A fundamental tension exists between fairness constraints and recommendation utility — enforcing minimum exposure for underperforming groups necessarily diverts selections from higher-reward arms, increasing cumulative regret. This fairness-utility tradeoff is a central challenge in fair bandit design and a key evaluation dimension of this thesis.

Huang et al. [24] propose FairLinUCB, a fair contextual bandit framework that addresses this tradeoff through two complementary mechanisms. The framework partitions the arm set into m demographic groups $\{G_1, G_2, \dots, G_m\}$, assigns each group a predefined target exposure rate L_i representing the minimum proportion of trials in which an arm from G_i must be selected, and tracks actual exposure $N_i(T)$ — the number of times arms from G_i have been selected up to trial T . The fairness constraint is satisfied when

$$\frac{N_i(T)}{T} \geq L_i \quad \forall i$$

The first mechanism is an exposure constraint that treats arm selection as a constrained optimization problem. When a group’s cumulative exposure falls below L_i , the algorithm adjusts its selection strategy to prioritize arms from that group, even if they carry slightly lower reward estimates, ensuring fairness constraints are satisfied over the horizon T while still aiming for high utility. **The second mechanism** is a reward discrepancy penalty, an in-processing score-augmentation approach that tracks the per-arm reward gap between demographic groups,

$$\Delta_a = \bar{r}_a^{s+} - \bar{r}_a^{s-}$$

where $\bar{r}_a^{s+}$ is the average reward for arm a given to the advantaged group and $\bar{r}_a^{s-}$ is the average reward for arm a given to the disadvantaged group and defines a fairness penalty,

$$F_a = -\text{sign}(\bar{r}_a^{s+} - \bar{r}_a^{s-}) \cdot \Delta_a$$

which is added directly to the UCB score. The sign function ensures the penalty is always applied in the direction that reduces disparity. Arms that treat demographic groups equally receive no penalty, while arms exhibiting large reward discrepancies are penalized proportionally — the more equitably an arm treats demographic groups, the smaller its penalty.

This thesis adopts the score-augmentation principle of Huang et al. [24]’s second mechanism and adapts it to the NeuralLinUCB framework described in Section 2.4. Specifically, since direct reward discrepancy tracking is not feasible in an offline simulation setting, the fairness signal is adapted to monitor recommendation rate gaps across gender and genre categories instead, and extended to operate over neural representations $\phi(\mathbf{x}; \mathbf{w}) \in \mathbb{R}^p$ rather than raw context vectors. This adaptation constitutes one of the primary contributions of this thesis and is formalised in Chapter 3.

2.6 Positioning and Novelty of this Thesis

This research is positioned at the intersection of deep representation learning and algorithmic ethics. By synthesizing the current state-of-the-art systems, this section describes the novelty of Fair-NeuralLinUCB and identifies the critical gap it addresses.

2.6.1 Theoretical Positioning: Efficiency and Ethics

Neural-based architectures have become the gold standard for recommendation systems due to their ability to map high-dimensional contextual feature matrices into latent spaces (z), thereby uncovering complex patterns that linear models overlook. While the efficiency of these models, particularly within Reinforcement Learning and Contextual Bandit frameworks, has been extensively demonstrated [61, 67], their “black-box” nature frequently neglects the socioeconomic and ethical implications of the recommendations they produce.

This study asserts that algorithmic fairness must be integrated into, rather than separated from, efficiency improvements. A system that delivers individualized “fair” solutions but lacks overall predictive accuracy fails to provide business value. Conversely, a highly efficient system that perpetuates demographic bias lacks ethical legitimacy. This research positions itself as an attempt to harmonize these two objectives, building a framework that is both economically viable and ethically grounded.

2.6.2 Identified Research Gap and Novelty

Despite significant advances in fairness research, a distinct gap remains: **no existing work successfully combines the non-linear feature extraction capabilities of Neural Contextual Bandits with active group-level fairness constraints.**

The positioning of this study is defined by three key factors: **Architectural Hybridity:** Previous studies have relied on either neural bandits for utility [53] or linear bandits for fairness [24]. To our knowledge, this work is among the first to integrate both approaches into a unified in-processing pipeline. **In-Processing Fairness:** Unlike post-processing methods that re-rank results after the model has already learned biased patterns, this study emphasizes in-processing mitigation, in which the reward function itself is regularized by a fairness penalty. **Societal Scalability:** Although this study utilizes movie recommendations as a tractable use case, the underlying logic is designed to address high-stakes domains such as medical or political recommendation systems, where algorithmic bias can have profound real-world consequences.

2.6.3 Methodological Limitations and Scope

In positioning this work, it is necessary to clarify the boundaries of its claims. This study focuses on Group Fairness (Demographic Parity). While demographic parity ensures an equitable distribution of genres across protected groups, it does not explicitly address Individual Fairness.

A recognized limitation of this study is that strict reliance on group-level demographic parity may inadvertently introduce a “fairness tax” at the individual level. This research is a critical step toward group-level equity, providing a foundation for future work on individual-level stereotype mitigation.

3

Methodology

This chapter presents the research strategy and methodology used in this study. We justify the choice of a Design Science Research approach and an experimental evaluation, critically review alternatives, and discuss ethical considerations. The methodology is situated within recent research on recommendation systems and algorithmic fairness. We also describe the data collection, analysis, and machine learning pipeline, as well as the software tools and their ethical implications.

3.1 Research Strategy and Method

3.1.1 Research Framework

This study adopts a **Design Science Research (DSR)** framework, specifically a Development- and Evaluation-Focused DSR approach, to construct and evaluate the Fair-NeuralLinUCB algorithm. The purpose of DSR is to design innovative artifacts and to systematically investigate them and their interactions with the target environment [27]. This approach focuses on developing an artifact that addresses a practical research problem and, through rigorous empirical assessment, contributes new knowledge to the scientific base [23]. Within this framework, an experimental simulation

strategy is appropriately used to assess the causal impact of the fairness-constrained neural algorithm relative to established baselines.

This research is a successive study building upon current state-of-the-art recommendation architectures. It specifically integrates the Neural-LinUCB model's [53] nonlinear representational power with Fair-LinUCB's [24] fairness constraints. While Neural-LinUCB approximates nonlinear reward functions, it ignores demographic disparities. Conversely, Fair-LinUCB mitigates algorithmic bias but is limited by its linear assumptions. The resulting Fair-NeuralLinUCB algorithm addresses both utility and demographic equity, offering a novel solution to the utility-fairness trade-off at the forefront of recommendation system research.

Following Johannesson and Perjons' design science framework [27], the research is structured into five activities: **(1) Explicate Problem** - addressing the systemic decoupling of utility optimization and fairness constraints in high-dimensional recommender systems; **(2) Define Requirements** - the Fair-NeuralLinUCB must provide neural representational power, active demographic bias mitigation, and computationally scalable regret updates; **(3) Design and Develop Artifact** - combining a neural feature extractor with a constrained LinUCB-based decision layer; **(4) Demonstrate Artifact** - operationalizing the model using a simulation on MovieLens 1M dataset to observe recommendation and fairness metrics; **(5) Evaluate Artifact** - Quantitatively benchmarking against Neural-LinUCB (for utility) and Fair-LinUCB (for fairness) across cumulative regret and demographic parity metrics.

3.1.2 Research Strategy

Under DSR - Demonstrate Artifact stage, the chosen strategy for testing and evaluating the designed artifact is **Model-Based Computational Simulation**. A simulation is an operational imitation of the behavior of a real-world process or system over time [27]. Algorithm design frames the balance between utility and fairness, while simulation enables controlled, reproducible testing of these dynamics [23].

In this study, the simulation environment leverages a real-world benchmark dataset, **MovieLens 1M** [22], obtained via secondary data collection, and employs a **Reward Oracle** approach to mimic a dynamic real-world

user environment. To conduct a rigorous comparative analysis, the simulation executes three distinct models under identical hyperparameter configurations, ensuring that variations in cumulative regret and demographic fairness are attributable solely to architectural design.

3.1.3 Alternative Research Strategy

Guided by the established literature on similar algorithmic problems, several alternative evaluation strategies were systematically assessed.

First, a formal **mathematical and logical proof** strategy was considered to derive a theoretical regret analysis, following the precedent set by the foundational papers introducing the LinUCB and Fair-LinUCB algorithms [35, 24]. This strategy was ultimately disregarded for two reasons: these foundational studies focus on introducing a new algorithm optimized for worst-case adversarial scenarios rather than addressing the specific engineering nuances of recommendation systems, whereas this study aims to build a model optimized for practical, average-case deployment. Furthermore, the integration of a deep neural network introduces complex, non-linear representation learning that requires highly restrictive, unrealistic assumptions for a formal mathematical proof, whereas a standard linear bandit operates on a static feature space where proofs are straightforward.

Second, a **qualitative case study** strategy, involving interviews and focus groups, was considered to explore users' perception of fairness. This strategy was rejected because qualitative approaches cannot verify mathematical convergence or calculate cumulative regret, both of which require significant statistical power [14]. Additionally, human judgments of algorithmic fairness are frequently compromised by self-interest and cognitive biases [58]. To train and validate an autonomous decision-making system, metrics must rely on objective, quantifiable definitions (e.g., Demographic Parity)—a requirement that can only be fulfilled through quantitative methods.

Finally, a **live A/B testing** strategy in a real-world production environment was evaluated, similar to the study by Shi et al. [53]. While field experimentation is the gold standard for user evaluation, deploying an unverified algorithm to test bias mitigation is ethically problematic because it

requires deliberately exposing a subset of real users to algorithmic disparities. Moreover, executing a live pipeline at the scale necessary for contextual bandit convergence requires significant computational infrastructure and operational costs that were outside the feasible scope of this study.

3.1.4 Data Collection Method

The MovieLens 1M [22] dataset is used to validate the algorithm. This dataset is a secondary dataset that benchmarks real-world user behavior and is widely used in recommendation system research. Synthesizing a biased dataset is considered an alternative data collection method [41]. It's not viable enough to replicate real-world human behavior, including demographic distribution. Other alternative secondary datasets considered include the Last.fm dataset and the Open-source YouTube dataset. The Last.fm dataset represents implicit feedback (play counts) rather than explicit satisfaction (ratings), making reward binarization less clean for standard bandit testing. Open-source YouTube datasets are highly anonymized, heavily fragmented, and often lack verified, explicit user demographic metadata (such as gender) required to strictly evaluate demographic parity bounds.

Movie Lens 1M contains explicit, clean historical ratings combined with verified self-reported demographic metadata (gender, age, occupation), making it uniquely suited for testing demographic parity. Gender-based disparities in recommendation quality have been empirically documented within the MovieLens 1M dataset [17]. This provides direct motivation for the fairness intervention proposed in this thesis.

3.1.5 Simulation Environment

The simulation environment comprises an SVD Reward Oracle that represents the real-world reward landscape. This was chosen over the alternative offline log replay method introduced by Li et al. [35]. Unlike standard offline log replaying, which is constrained by data sparsity and high variance (especially given that the MovieLens 1M dataset exhibits a highly sparse rating density of approximately 4%), Reward Oracle uses a pre-trained SVD model to estimate latent user and item features from the historical data. The resulting matrix serves as a ground-truth environment, supplying a continuous, low-variance reward signal for the agent's actions [48].

Demographic Parity was selected as the primary metric for mitigating algorithmic bias. Alternative fairness paradigms, such as Equal Opportunity, which requires observing post-consumption satisfaction metrics across all groups, or Individual Fairness, which necessitates measuring fine-grained, user-level stereotyping biases, were rejected due to constraints imposed by the simulated bandit framework. Demographic Parity, by contrast, focuses on the statistical equality of recommendation exposure, offering an objective and quantifiable optimization constraint for sequential decision-making.

Gender was chosen as the protected sensitive attribute over alternative meta-data attributes such as age or occupation due to two primary factors. First, the gender distribution in the MovieLens 1M dataset is heavily imbalanced, with a disproportionately high male-to-female ratio that constitutes a critical algorithmic bias. Second, gender provides a clean, binary baseline. In contrast, attributes like age and occupation exhibit high cardinality, which can fragment the bandit’s internal data splits and dilute the fairness function’s statistical power during the critical early exploration phases. More complex intersectional fairness attributes were also excluded as they fell outside the operational scope of this study.

3.1.6 Data Analysis Method

The data analysis method applies a multi-dimensional **quantitative** framework to evaluate the simulation logs across three distinct models. Performance is analyzed using cumulative reward and cumulative regret to verify sublinear regret growth, alongside the Hit Rate, which measures the proportion of recommendation steps in which the agent suggests an item that exceeds a baseline user satisfaction threshold (i.e., the proportion of positive ratings among the total recommendations made by the agent during the simulation loop). Simultaneously, algorithmic fairness is audited by tracking the gender-based exposure gap (the real-time divergence in the genres recommended to male versus female users). To ensure consistency and validity, the fairness penalty parameter was set according to the recommended configurations established by Huang et al. [24], ensuring that the artifact’s trade-off between utility and demographic parity aligns with established benchmarks.

Traditional static metrics, specifically Precision and Recall, were deliberately excluded from this analysis due to structural incompatibilities with sequential learning. These metrics require a frozen, fully observed test set; however, because the MovieLens 1M dataset has a sparse 4% rating density, an exploring bandit would be incorrectly penalized for false positives whenever it recommends a highly relevant but historically unrated item. By bypassing static information retrieval metrics in favor of cumulative tracking and a dynamic Hit Rate powered by the SVD Reward Oracle, the analysis accurately evaluates the architecture’s online exploration trajectory and group-equity trade-offs without being skewed by historical data omissions.

3.1.7 Ethical Consideration

The ethical challenges in this study focus on Algorithmic Accountability and mitigating systemic demographic discrimination. Algorithmic Accountability refers to the responsibility of system designers to prevent and address harm arising from discriminatory algorithmic outcomes [10]. Historical biases in training data can lead to “bias amplification,” where sequential recommendation systems inadvertently reinforce and scale existing societal inequalities [39].

The choice of an in-processing fairness method is itself an ethical decision. By integrating group-level exposure constraints directly into the Fair-NeuralLinUCB learning process, the study prioritizes equity at the point of decision-making rather than correcting bias post hoc, ensuring that the protected user group receives a comparable level of recommendation utility to the advantaged user group throughout the simulation.

Furthermore, this research actively navigates the Fairness-Utility Trade-off by adhering to the validated fairness configurations established in the baseline literature, balancing recommendation optimization against structural group-equity mandates [24]. Finally, in accordance with data ethics standards, the MovieLens 1M dataset is handled in its fully anonymized form to protect user privacy, ensuring the system remains a responsible, reproducible, and accountable artifact of design research.

3.2 Application of Strategy and Method

Building the Fair-NeuralLinUCB model (artifact) and the experimental evaluation setup (simulation) is implemented through a structured machine learning pipeline, ensuring observed improvements in fairness or regret are directly attributable to the Fair-NeuralLinUCB architecture, not data fluctuations [14]. The pipeline is implemented in four distinct stages.

3.2.1 Data Pre-processing and Contextualization

The MovieLens 1M dataset [22], comprising over one million timestamped ratings (1 to 5) on 3,900 movies by 6,040 users, with rich metadata (age, gender, occupation, genre, etc.), serves as the empirical foundation for the simulation. To ensure the stability and validity of the results, the data underwent a three-stage pre-processing pipeline:

Filtering and Quality Control: Only movies with more than 100 ratings and users with more than 50 ratings were retained. This threshold is a mechanical necessity; sparse data points often lack the representational density required for deep learning, leading to numerical instability and exploding gradients during backpropagation in the Neural-LinUCB component. While thresholds such as restricting the environment to top-N rated items or high-activity users introduce popularity bias, this approach prioritizes a high-volume, high-value data stream over stochastic noise.

Encoding and Reward Binarization: Feature engineering was applied to transform qualitative attributes into a format suitable for neural computation:

- **User Metadata:** Categorical variables (Gender, Occupation, and Age groups) were encoded using One-Hot Encoding.
- **Movie Genres:** Because movies can belong to multiple categories, Multi-Hot Encoding was used to represent genre diversity accurately.
- **Reward Mapping:** Ratings were converted into a binary reward system where scores of 4 – 5 map to 1 (Positive Feedback) and scores of

1 – 3 map to 0 (Negative/Neutral Feedback). This binarization effectively translates the static rating matrix into the binary reward signal required for the Bernoulli-style bandit model.

Joint Context Construction: Encoded user and movie features were combined into 48-dimensional input vectors for the neural network, enabling learning of complex interactions.

3.2.2 Reward Oracle Construction

The second stage develops a **Reward Oracle**, which serves as the ground-truth environment for the online learning simulation. Because contextual bandits require reward feedback for every recommended action, and the original MovieLens 1M logs cover only 4.26% of all possible user-movie interactions, the oracle provides counterfactual reward signals for user-movie pairs absent from the original logs. Matrix Factorization (using the Surprise library) was used to predict missing interactions. FunkSVD minimized a regularized squared error on observed ratings (with SGD), making it effective for MovieLens 1M’s sparse data. Decomposing ratings into $k = 50$ latent factors, let the oracle estimate ratings for “unwatched” movies using the dot product of user and item vectors. The construction of the oracle followed a rigorous five-step pipeline, ensuring the simulation environment remained stable and accurate.

1. **Dataset Integration:** The filtered ratings dataset was loaded into the Surprise library. This ensures the oracle operates on the same data as the rest of the pipeline. To provide SVD with a richer signal, the original ratings (1 – 5) were used instead of binary ratings.
2. **Model Training:** The SVD model was trained using 50 latent factors over 20 epochs with a learning rate of 0.005, and an $L2$ regularization of 0.02. These parameters were selected by manual tuning to prevent overfitting while capturing sufficient variance in user behavior. The oracle construction used a fixed random seed (42) to ensure the reward environment remains constant across all experiments.
3. **Validation:** A held-out validation set (10% of observed ratings) was used to assess the model’s accuracy. The oracle’s ability to predict

ratings was verified before it was used as the ground-truth reward environment for the simulation.

4. **Dense Matrix Reconstruction:** The trained model was used to predict a full, dense rating matrix for all users across all candidate movies.
5. **Reward Binarization:** To align with the Bernoulli bandit framework, the predicted continuous ratings were binarized. Using the same threshold as the pre-processed data, any predicted rating ≥ 4 was mapped to a reward of 1, and ratings < 4 were mapped to 0.

The binary matrix creates a deterministic environment — each model receives an identical, pre-calculated reward from the oracle for a given user-movie pair it selects. With a fixed random seed and identical candidate sampling across all three models, this design ensures that observed performance differences are attributable solely to algorithmic design choices rather than to stochastic variation in the simulation environment.

3.2.3 Neural Representation Learning (ϕ)

Architecture: Feature extraction used a DNN to map 48-dimensional context vectors to a 64-dimensional latent space using a funnel architecture:

$Input(48) \rightarrow Linear(128) \rightarrow BN \rightarrow ReLU \rightarrow Dropout(0.1) \rightarrow Linear(64) \rightarrow BN \rightarrow ReLU \rightarrow Dropout(0.1) \rightarrow Output(1)$

Non-Linearity and Stability: ReLU activation function was employed to introduce the non-linearity required to capture complex user-item interactions that standard linear models fail to identify. Batch Normalization (BN) was placed before the activation to stabilize the input distribution, mitigate the vanishing gradient problem, and enable faster convergence.

Regularization via Dropout: To ensure the learned representations were robust and not merely “memorized” from the training set, Dropout ($p = 0.1$) was applied after each hidden layer. By randomly deactivating 10% of the neurons on each training pass, the network is forced to learn generalizable, redundant feature patterns, thereby significantly reducing overfitting.

Dimensionality Bottleneck: The transition from 128 to 64 neurons acts as an information bottleneck. This forces the network to distill the input

into a dense representation ($z = \phi(x)$), retaining only the features with the highest predictive power for rewards. A two-layer depth provides sufficient model capacity while maintaining the structural simplicity to prevent high variance in the final embeddings.

Data Partitioning: The training dataset is constructed by concatenating the user and movie features from the filtered historical rating logs to form the input feature space. The discrete binary reward from the Reward Oracle matrix defines the corresponding target variable. To test generalization, the dataset was split into Training (80%), Validation (10%), and Test (10%) sets. A Stratified Shuffle Split (via scikit-learn with random seed 42) was used to maintain a consistent positive-to-negative reward ratio across all subsets, preventing demographic or rating imbalances from skewing the results. The training set was used to optimize the Neural Feature Extractor’s weights, while the validation set was used for hyperparameter tuning and early stopping to mitigate overfitting. Finally, performance metrics were established on the test set before the network weights were frozen, ensuring a stable latent representation (z) for the subsequent 50,000-step bandit simulation.

Loss Optimization: To mitigate class imbalance, a weighted BCEWithLogitsLoss scaled by inverse class frequency was utilized to prevent the model from developing a predictive bias toward the majority class.

Hyperparameter Tuning and Optimization: Manual tuning of the learning rate, dropout, and architecture was conducted to minimize validation loss, utilizing the Adam optimizer with an initial learning rate of $1e-3$ (later refined to $5e-4$) and a batch size of 256. To further prevent overfitting and ensure convergence, ReduceLROnPlateau (patience: 2, factor: 0.5) was used to manage learning rate plateaus, while an Early Stopping mechanism (patience: 5) halted Training once the validation loss ceased to improve.

Freezing the Representation: After optimal Validation, the output layer was removed, and hidden layer weights were frozen. This made the DNN a static feature extractor, allowing Fair-NeuralLinUCB to operate in a stable 64-dimensional latent space with consistent features during simulation.

3.2.4 Online Learning Simulation Design

The last stage is a large-scale online learning simulation to assess the trade-off between recommendation accuracy and demographic parity. The simulation mimics real-world sequential learning of user preferences. The simulation environment uses key parameters to ensure computational feasibility and statistical significance:

- **Statistical Robustness** ($10RandomSeeds$): Each simulation is repeated across 10 distinct random seeds to account for stochastic variance in the bandit setting, controlling for the influence of user sampling order and candidate set randomness. Using identical seeds across all three models ensures that observed performance differences are attributable to algorithmic design rather than sampling variation [33].
- **Time Horizon** ($T = 50,000$): The agent takes 50,000 sequential recommendation steps, providing each of the candidate arms with sufficient appearances in the candidate set for the LinUCB covariance matrices to accumulate reliable reward estimates beyond the initial exploration phase.
- **Candidate Sampling** ($K = 200$): At each step, 200 candidate movies are randomly sampled, balancing computational feasibility with action space coverage. This reflects deployment constraints in real recommendation systems, where scoring all available items at each request is impractical.
- **Feedback Loop**: The agent receives binary feedback exclusively for the selected arm — the SVD oracle reward for all unselected candidates remains unobserved, implementing the partial observability characteristic of the contextual bandit setting [35].

Comparative Model Architecture: Three models were implemented to isolate the effects of neural representations and fairness constraints (see Table 3.1). Their performance was evaluated through pairwise comparisons.

The learning head uses $L2$ regularization ($\lambda = 1.0$) to initialize the covariance matrix as an identity matrix, ensuring invertibility and high initial UCB variance. This encourages exploration of diverse genres early on. A sensitivity analysis tuned the exploration-exploitation parameter α over a range of values from 0.001 to 1.0.

Table 3.1: Structural Comparison of Experimental Bandit Architectures

Feature	Neural-LinUCB	Fair-LinUCB	Fair-NeuralLinUCB
Input Context (x)	Raw 48-dim Vector	Raw 48-dim Vector	Raw 48-dim Vector
Feature Mapping	Neural (ϕ)	Identity (Linear)	Neural (ϕ)
Latent Space (z)	64-dim (Dense)	48-dim (Sparse)	64-dim (Dense)
Fairness Tracker	Disabled	Active	Active
Exploration (α)	0.05	0.05	0.05
Penalty Mechanism	None	Genre-Gap Penalty	Genre-Gap Penalty
Primary Goal	Maximize Utility	Balance Regret/Fairness	Balance Regret/Fairness

- Low Exploration ($\alpha < 0.1$): The model became greedy, converging too fast on sub-optimal choices, increasing long-term regret.
- High Exploration ($\alpha > 0.5$): The agent sampled too many low-reward movies, reducing cumulative reward and destabilizing fairness.
- Optimal ($\alpha = 0.05$): This value balanced exploration and accuracy for efficient learning.

The Fairness Tracker and Penalty Mechanism is a mechanical intervention designed to mitigate demographic disparity across different movie genres. It operates by monitoring the Recommendation Rate Gap by gender (gap_g) for every genre g . The gap is monitored after an initial run of at least 10 movies per gender. The logic follows a feedback-control loop.

1. **Monitoring:** The tracker calculates the recommendation rate of a genre g for each gender: $rec_rate_{g,gender} = \frac{count_{g,gender}}{total_{gender}}$.
2. **Detection:** A gap is identified if $|rec_rate_{g,Male} - rec_rate_{g,Female}| > \epsilon$ where ϵ is a set threshold.
3. **Intervention:** If a gap exceeds the threshold ϵ , a Fairness Bonus (or penalty) β is added to the UCB score for the underserved group.

$$B_a = \begin{cases} \beta \cdot (gap_g - \epsilon) & \text{if user} \in F \text{ and } gap_g > \epsilon \\ \beta \cdot (-gap_g - \epsilon) & \text{if user} \in M \text{ and } gap_g < -\epsilon \\ 0 & \text{otherwise} \end{cases}$$

In the Fair-NeuralLinUCB model, the final selection score for an arm a is calculated as:

$$Score(a) = \underbrace{\hat{\theta}^T z_a}_{\text{Exploitation}} + \underbrace{\alpha \sqrt{z_a^T A^{-1} z_a}}_{\text{Exploration}} + \underbrace{B_a}_{\text{Fairness Bonus}}$$

Where B_a is the dynamic adjustment provided by the tracker, this forces the model to explore “fairer” actions that it might otherwise ignore due to historical biases in the data. To enforce demographic parity, a threshold $\epsilon = 0.05$ is set for genre exposure disparity. If exceeded, a Fairness Bonus ($\beta = 2.0$) is applied to underserved arms’ UCB scores, pushing the policy toward the Fairness-Utility Pareto Frontier. A higher bonus increases fairness enforcement. Rather than adjusting these variables through an empirical sensitivity sweep, both the threshold (ϵ) and the bonus (β) were explicitly adopted from the recommended hyperparameter configurations validated in the foundational work of Huang et al. [24], ensuring that the agent’s exploratory constraints align precisely with established group-fairness benchmarks.

A fixed random seed ensured identical sampling of the $K = 200$ candidate arms across all models at each simulation step, enabling a direct comparison of the “Fairness Tax” without the interference of sampling noise.

3.2.5 Evaluation and Benchmarking Framework

Evaluation was conducted using a real-time logging and benchmarking system integrated within the simulation loop, enabling simultaneous monitoring of convergence and demographic alignment over the $T = 50,000$ horizon.

Learning efficiency was assessed through automated logging at 1,000-step intervals. Cumulative Regret (R_T) measured the opportunity cost between optimal and actual rewards, confirming sublinear convergence. Cumulative Reward logs and hit rate tracked total user satisfaction for each model.

The performance of the three models was benchmarked using statistical significance analysis of three key metrics: reward, regret, and gender exposure gap. For pairwise model comparisons, the paired t -test [43] served as the primary parametric method. To validate the normality assumption,

the Shapiro-Wilk test [52] was applied to paired differences, as visual inspection is unreliable with only 10 seeds. When the assumption of normality was violated, the Wilcoxon signed-rank test [59] was used as a robust nonparametric alternative. The Wilcoxon test, which operates on the ranks of absolute differences, is resilient to outliers and effective for small sample sizes where normality tests may lack statistical power. To address the limitation that statistical significance does not indicate effect magnitude, Cohen’s d effect size is reported alongside p -values for all comparisons.

The results produced by this experimental setup, including comparative regret curves and Pareto frontier analyses, are presented and discussed in Chapter 4.

3.2.6 Software Environment and Ethical Considerations

The experimental pipeline was built in Python 3.10 for efficiency and reproducibility. PyTorch is used for neural network development, and NumPy and Pandas for data operations. Scikit-learn was used to compute metrics such as AUC and to conduct sensitivity analyses. Other libraries used include Surprise for constructing reward oracles, Seaborn, and Matplotlib for visualizing learning curves and fairness trade-offs. All simulations were run in Jupyter Notebooks in VS Code with NVIDIA RTX 5070 GPU acceleration, ensuring consistent and reproducible results.

In accordance with university ethical guidelines, generative AI (Gemini and Claude) was used only for brainstorming, refining prose, and synthesizing the literature. All algorithms, math, and interpretations are the author’s original work. AI was used to clarify presentation, not create research content.

The research uses the MovieLens 1M open-source data [22] in compliance with the GroupLens license, keeping data anonymized and users unidentifiable. The study also serves as an ethical intervention, aiming to reduce filter bubbles and demographic disparities through the Fairness Tracker, thereby promoting more equitable, transparent AI.

4

Results

This chapter presents the empirical findings of the 50,000-step online simulation. The analysis evaluates the Neural-LinUCB, Fair-LinUCB, and Fair-NeuralLinUCB models across three dimensions: computational efficiency, predictive accuracy, and demographic parity.

4.1 Exploratory Data Analysis and Pre-processing Results

The dataset used for the final simulation consisted of a filtered subset of MovieLens 1M, where movies with more than 100 ratings and users with more than 50 ratings were retained. After initial filtering to ensure sufficient learning signals for the Bandit agents, the dataset characteristics were summarized. Table 4.1 presents the rating density of the environment before and after filtering, while Table 4.2 details the sparsity and distribution of ratings across the user and item sets before and after filtering.

As evidenced by 4.26% Raw Rating Density, the interaction matrix is highly sparse. The distribution of ratings per user exhibits a high standard deviation ($\sigma = 192.75$). This variance necessitates filtering and using the Reward Oracle (SVD) to provide counterfactual feedback, as the limited historical logs for the bottom 50% of users (Median = 96) would otherwise cause a

Table 4.1: Global Dataset and Simulation Environment Statistics

Metric	Value
Total Unique Users	6,040
Total Unique Movies	3,883
Total Unique Genres	18
Total Ratings	1,000,209
Rating Density	4.26%
<i>Post-Filtering for Simulation</i>	
Users with > 50 Ratings	4,297
Movies with > 100 Ratings	2,019
Rating Density	10.24%

Table 4.2: Comparative Descriptive Statistics: Raw vs. Filtered Dataset

Statistic	Ratings per User		Ratings per Movie	
	Raw	Filtered	Raw	Filtered
Count (n)	6,040	4,297	3,883	2,019
Mean (μ)	165.60	206.65	269.89	439.80
Std. Dev (σ)	192.75	184.69	384.05	390.50
Minimum	20.00	24.00	1.00	89.00
Median (50%)	96.00	140.00	123.50	300.00
Maximum	2,314.00	1,568.00	3,428.00	2,758.00

cold-start bottleneck for the Neural-LinUCB feature extractor. A comparative analysis of the descriptive statistics (Table 4.2) reveals the impact of the filtering process. By removing cold-start entities, the median number of ratings per movie increased from 123.50 to 300.00, a 143% improvement. Similarly, the mean ratings per user rose from 165.60 to 206.65. This shift is mathematically significant for the Neural-LinUCB model; it ensures that the latent feature extraction layer is trained on a more consistent, denser signal, reducing variance in reward estimation during the early phases of the simulation.

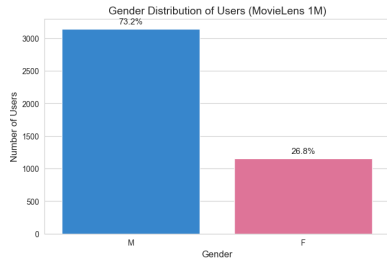
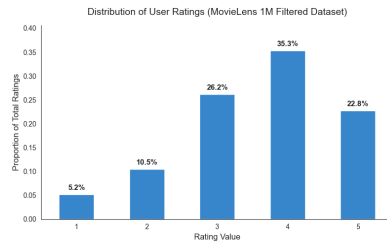
(a) Gender Distribution of the Filtered Dataset ($n = 4,297$).(b) Ratings Distribution of the Filtered Dataset ($n = 4,297$)

Figure 4.1: Summary of Dataset Skewness and Imbalance

The filtering process yielded a final simulation environment comprising 4,297 users and 2019 movies. As shown in Figure 4.1a, this subset exhibits a significant demographic imbalance with a male-to-female ratio of 2.73:1. This represents a higher degree of skewness compared to the raw dataset, suggesting that the most active reviewers in the MovieLens repository are disproportionately male. From a systems science perspective, this imbalance creates a high risk of ‘majority-class optimization,’ where the bandit agent could achieve low global regret by ignoring female preferences entirely. This finding provides a strong empirical justification for implementing the Fairness Tracker and the Genre-Gender Exposure Gap constraint.

Figure 4.1b illustrates the rating distribution within the filtered dataset. While positive ratings (≥ 4) represent 58.1% of the total interactions, the remaining 41.9% are distributed across neutral and negative feedback (1 – 3). To ensure the Neural Network feature extractor does not develop a ‘positivity bias’—where it over-predicts rewards simply by favoring the majority class, it is trained on observed user-movie interactions with binary labels derived from the oracle matrix rather than raw rating thresholds (reducing positive ratings to 32%). This ensures that the model remains sensitive to both positive and negative user signals, which is mathematically essential for computing accurate regret in subsequent bandit simulations.

4.2 Model Components Performance

4.2.1 Reward Oracle Reliability

Because offline recommendation simulation cannot interact with users in real time, the SVD Reward Oracle serves as a computational surrogate. The model demonstrated high-fidelity estimation with an RMSE of 0.8498 (exceeding the 0.90 target), a MAE of 0.6668, and an AUC of 0.8172 (significantly exceeding the 0.72 target). The resulting Test Accuracy of 0.65 provides a robust benchmark for the simulation. With a positive reward rate of 20.2%, the Oracle creates a realistic environment where a random policy yields a baseline success rate of approximately 20%. This specific reward density ensures a meaningful learning signal for the bandit agents, as it is high enough to enable convergence yet sparse enough to require sophisticated exploration.

Table 4.3: Classification Report for the SVD Reward Oracle

Class	Precision	Recall	F1-Score	Support
Disliked (0)	0.55	0.91	0.68	36,838
Liked (1)	0.88	0.46	0.61	51,959
Macro Avg	0.71	0.69	0.65	88,797
Weighted Avg	0.74	0.65	0.64	88,797

The classification report (Table 4.3) further illustrates Oracle’s reliability as a reward generator at the binarized threshold. Precision of 0.88 for Liked class means the reward given by the oracle for a given bandit arm is accurate 88% of the time, preventing bandit learning from “false positives”. Although the Recall (0.46) is lower, this effectively models a “strict” environment in which rewards are not given easily. This forces the Bandit to find truly high-confidence matches to maximize utility. The Macro Average F1-score (0.65) is significantly high. This is ideal since the Oracle is more accurate than the Bandit agent; this gap provides the necessary room for the agent to learn and improve.

4.2.2 Neural Feature Extractor

The trained Neural Network (NN) achieved a Test AUC of 72.00%, a Test Accuracy of 69.00%, and a Test Loss of 0.45. These metrics successfully satisfy the established performance benchmarks (> 0.72 and > 0.65 , respectively). The significant reduction in Test Loss to 0.45 indicates superior convergence and a more refined mapping of the high-dimensional MovieLens input into the 64-dimensional latent space (z).

The performance of the binary classifier determines the quality of the latent representation z passed to the Bandit. Table 4.4 breaks down the effectiveness of the model across the two target classes: Disliked (0) and Liked (1).

A key achievement in this final model is the balance between the two classes. The model maintains a high Precision of 0.77 for the “Disliked” class, while the Precision for the “Liked” class is 0.52. Given that the Liked class constitutes only 32% of the test set, this represents a substantial improvement over a frequency-based random classifier, demonstrating that the feature extractor has learned meaningful signal beyond majority-class prediction.

Table 4.4: Classification Report for the Neural Feature Extractor

Class	Precision	Recall	F1-Score	Support
Disliked (0)	0.77	0.78	0.77	60,412
Liked (1)	0.52	0.51	0.51	28,385
Accuracy			0.69	88,797
Macro Avg	0.64	0.64	0.64	88,797
Weighted Avg	0.69	0.69	0.69	88,797

The proximity of the Macro and Weighted F1-scores at 0.64 and 0.69, respectively, suggests that the feature extractor has substantially mitigated majority-class bias. By providing a latent representation $\phi(x)$ that captures meaningful signals for both positive and negative user feedback, the model offers a robust foundation for the Neural Bandit models. The Bandit can now leverage these optimized features to focus its exploration on the more nuanced boundaries of user preference, rather than struggling with initial classification noise in the raw feature space.

4.3 Comparative Performance: Utility and Gender Fairness

The primary objective of this study was to evaluate the trade-off between recommendation utility, measured by the Regret, and demographic fairness, quantified by the Gender Gap. Table 4.5 presents the comparative performance summary for the three models, with results averaged over 10 independent random seeds, each consisting of a 50,000-iteration simulation.

Table 4.5: Performance Comparison (Mean $\pm$ SD)

Metric	NeuralLinUCB	FairLinUCB	Fair-NeuralLinUCB
Total Reward	36839.2 $\pm$ 425.2	35455.4 $\pm$ 1031.4	37107.2 $\pm$ 477.3
Total Regret	11722.2 $\pm$ 399.1	13106.0 $\pm$ 1048.8	11454.2 $\pm$ 486.6
Gender Gap	0.0270 $\pm$ 0.0054	0.0129 $\pm$ 0.0082	0.0217 $\pm$ 0.0073
Hit Rate (All)	0.7368 $\pm$ 0.0085	0.7091 $\pm$ 0.0206	0.7421 $\pm$ 0.0095
Hit Rate (M)	0.7440 $\pm$ 0.0088	0.7126 $\pm$ 0.0217	0.7480 $\pm$ 0.0104
Hit Rate (F)	0.7170 $\pm$ 0.0090	0.6997 $\pm$ 0.0186	0.7262 $\pm$ 0.0091

Fair-NeuralLinUCB achieves the highest reward and incurs the least regret, outperforming both baselines in utility. The large standard deviation of FairLinUCB (± 1031) compared to Fair-NeuralLinUCB (± 477) indicates FairLinUCB is less stable across seeds. FairLinUCB achieves the smallest gender gap but at the cost of substantially lower reward and higher regret. Fair-NeuralLinUCB sits between the two — fairer than NeuralLinUCB but not as fair as FairLinUCB. Fair-NeuralLinUCB also achieves a higher hit rate for both genders than NeuralLinUCB does.

To determine the statistical validity of these findings, a series of significance tests was conducted. The following three tables 4.6, 4.7, and 4.8 detail pairwise comparisons between models, evaluating Cumulative Reward, Cumulative Regret, and the Gender Gap to determine whether the observed performance differences are statistically significant.

Table 4.6: Significance Test Results: NeuralLinUCB vs. Fair-NeuralLinUCB

Metric	Statistical Test	<i>p</i>-value	Cohen's <i>d</i>	Sig. ($\alpha = 0.05$)
Total Reward	Paired <i>t</i> -test	0.2445	-0.394	n.s.
Total Regret	Paired <i>t</i> -test	0.2445	0.394	n.s.
Gender Gap	Paired <i>t</i> -test	0.0256	0.844	Yes (*)

Fair-NeuralLinUCB reduced the gender stereotype gap significantly ($p = 0.026$, $d = 0.844$) without a statistically significant performance cost, with reward and regret differences consistent with random variation ($p = 0.245$).

Table 4.7: Significance Test Results: FairLinUCB vs. Fair-NeuralLinUCB

Metric	Statistical Test	<i>p</i>-value	Cohen's <i>d</i>	Sig. ($\alpha = 0.05$)
Total Reward	Paired <i>t</i> -test	0.0031	-1.266	Yes (**)
Total Regret	Paired <i>t</i> -test	0.0031	1.266	Yes (**)
Gender Gap	Paired <i>t</i> -test	0.0524	-0.706	n.s.

Fair-NeuralLinUCB significantly outperformed FairLinUCB on reward and regret ($p = 0.003$, $d = 1.266$) while achieving statistically comparable gender gap reduction ($p = 0.052$, $d = 0.706$), demonstrating that the neural representation substantially improves utility without sacrificing fairness.

NeuralLinUCB significantly outperformed FairLinUCB on all utility metrics ($p = 0.002$, $d = 1.114$) while FairLinUCB significantly reduced the gender gap ($p = 0.001$, $d = 1.497$), confirming the utility-fairness tradeoff that Fair-NeuralLinUCB is designed to address.

Table 4.8: Significance Test Results: NeuralLinUCB vs. FairLinUCB

Metric	Statistical Test	p -value	Cohen's d	Sig. ($\alpha = 0.05$)
Total Reward	Wilcoxon Signed-Rank	0.0020	1.114	Yes (**)
Total Regret	Wilcoxon Signed-Rank	0.0020	-1.114	Yes (**)
Gender Gap	Paired t -test	0.0011	1.497	Yes (**)

Figure 4.2 illustrates the comparative performance of the bandit architectures across three key metrics. The cumulative reward and regret trajectories are plotted with 95% confidence intervals (CIs), representing the variance across 10 independent simulation seeds.

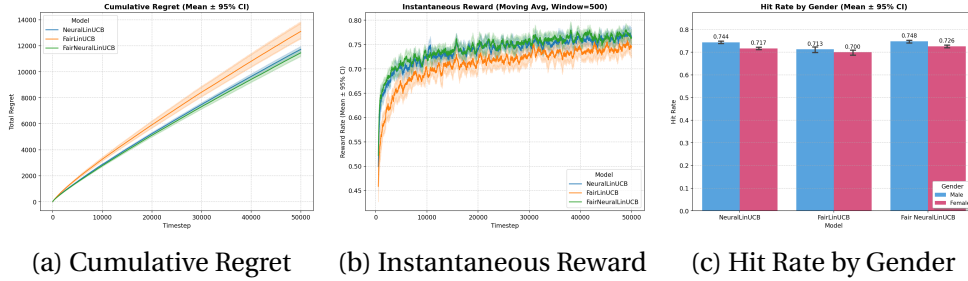


Figure 4.2: Performance Analysis of Bandit Architectures.

In the Cumulative Regret analysis 4.2a, FairLinUCB shows a steeper, more linear growth rate with a noticeably widening 95% Confidence Interval (CI) over time, indicating increasing instability and a higher opportunity cost. In contrast, the NeuralLinUCB and Fair-NeuralLinUCB curves nearly overlap, maintaining much tighter CIs and demonstrating superior sublinear convergence. This trend is mirrored in the Instantaneous Reward plot 4.2b, where both neural models converge to a higher reward ceiling than the linear baseline. Finally, the Hit Rate by Gender chart 4.2c confirms that while FairLinUCB yields the lowest overall performance across groups, it also maintains the smallest gap. The hybrid Fair-NeuralLinUCB successfully recovers the lost utility, achieving higher hit rates for both genders while maintaining a narrow disparity gap.

The experimental results, provide a robust validation of the research objectives. The performance of the proposed Fair-NeuralLinUCB is evaluated against the standard NeuralLinUCB and the linear FairLinUCB across both utility and fairness dimensions.

Utility: The Neural Advantage

As detailed in Table 4.5, Fair-NeuralLinUCB achieves the highest overall total reward and hit rate among the tested models. Notably, the difference in raw reward between the standard NeuralLinUCB and the fairness-constrained Fair-NeuralLinUCB is not statistically significant ($p = 0.245$; Table 4.6). This “null result” is expected and confirms that incorporating group-level exposure constraints into the neural framework does not meaningfully diminish predictive utility.

In contrast, the neural hybrid model outperforms the linear FairLinUCB. Table 4.7 reports a highly significant difference in both total reward and regret ($p = 0.003$) between FairLinUCB and Fair-NeuralLinUCB with a very large effect size (Cohen’s $d = 1.266$). As illustrated in Figures 4.2a and 4.2b, the neural models exhibit stable, sublinear regret growth and converge toward a higher reward ceiling, with their curves nearly overlapping. This underscores the strong predictive power of deep feature extraction.

Conversely, FairLinUCB fails to keep pace, plateauing at a lower reward ceiling and exhibiting greater regret variability over time. This suggests higher stochastic variance and reduced stability when rigid linear features are forced to satisfy fairness constraints. These results indicate that the “fairness tax”—the inevitable loss in utility required to achieve demographic parity—is substantially higher for traditional linear models. By leveraging a neural feature extractor to provide a more robust latent representation (z), the hybrid architecture effectively mitigates this tax, achieving parity without sacrificing the predictive accuracy of the unconstrained neural baseline.

Fairness: Mitigating Demographic Disparity

The evaluation of demographic parity reveals that the unconstrained NeuralLinUCB model exhibited the highest gender gap (0.027, Table 4.5), underscoring an inherent bias in recommendation quality between male and female cohorts. Fair-NeuralLinUCB significantly reduced this disparity to 0.0217, a statistically significant improvement ($p = 0.026$, $d = 0.844$, Table 4.6). Notably, it also improved the hit rate for female users (0.726 vs. 0.717, Table 4.5), indicating that fairness is achieved by increasing accuracy for the underserved group rather than lowering it for the majority.

The Hit Rate by Gender (Figure 4.2c) illustrates the tangible impact of these architectures. FairLinUCB yields the lowest overall hit rates for both male (0.713) and female (0.670) users, yet it achieves the narrowest absolute gap between the two groups. This suggests that while the linear model’s rigid constraints are highly effective at enforcing parity by strictly limiting exposure variance, they do so at the cost of significant “utility erosion.” Fair-NeuralLinUCB provides a superior balance, achieving higher hit rates (0.748 and 0.726, respectively) while maintaining a minimal gender gap comparable to the fair linear baseline (Table 4.7). These results confirm that integrating high-dimensional neural feature mapping enables the system to capture complex user-item interactions that linear models overlook. The resulting system is not only more equitable but also maintains significantly higher performance, effectively decoupling the traditional trade-off between fairness and utility.

The Comparative Frontier

While FairLinUCB achieved the narrowest overall gender gap (0.0129, Table 4.5), this parity was attained at the expense of the lowest utility across all performance metrics. The significant difference in the gender gap between NeuralLinUCB and FairLinUCB ($p = 0.0011$, Table 4.8) underscores a critical architectural trade-off: although linear models are more easily constrained to meet ethical requirements, they lack the representational capacity necessary to remain competitive in high-dimensional recommendation tasks.

Fair-NeuralLinUCB distinguishes itself as the superior hybrid, successfully bridging the gap between high-accuracy but biased neural baselines and fair but low-utility linear models. By providing a statistically validated “middle ground,” the hybrid architecture demonstrates that it is possible to prioritize both user satisfaction and ethical AI constraints simultaneously. This suggests that the latent representations provided by the neural feature extractor allow the model to navigate the fairness-accuracy Pareto frontier more effectively than traditional linear approaches, making it the most robust candidate for deployment in sociotechnically sensitive environments.

5

Discussion

5.1 Conclusion: Answering the Research Question

The primary objective of this study was to investigate how integrating group-level exposure constraints into a neural contextual bandit framework affects the trade-off between recommendation accuracy and fairness. Based on the comparative analysis in Chapter 4, the research questions are answered as follows:

- **RQ1: Does Fair-NeuralLinUCB reduce group-level stereotype bias versus NeuralLinUCB?** Yes. The Fair-NeuralLinUCB model successfully reduces the gender stereotype gap (from 0.0270 to 0.0217) compared to unconstrained NeuralLinUCB, a statistically significant improvement ($p = 0.026$, $d = 0.844$). This reduction comes at no statistically significant cost to utility — the difference in reward and regret between NeuralLinUCB and Fair-NeuralLinUCB is non-significant ($p = 0.245$). It confirms that the fairness penalty is absorbed by the richer latent representation.
- **RQ2: Does Fair-NeuralLinUCB maintain performance versus Fair-LinUCB?** Yes. Fair-NeuralLinUCB significantly outperformed the Fair-LinUCB model in both cumulative reward and regret ($p = 0.0031$, $d = 1.266$), while achieving a statistically comparable reduction in the

gender gap ($p = 0.052$), effectively reclaiming the “utility tax” typically associated with algorithmic fairness.

In summary, the neural feature extractor provides the representational capacity necessary to decouple the fairness-accuracy trade-off, enabling ethical constraints to be met without sacrificing system utility. The hybrid model achieves a superior position on the fairness-utility Pareto frontier compared to either constrained linear or unconstrained neural baselines. The result supports the thesis’s central claim that algorithmic fairness and predictive utility need not be treated as opposing objectives when a sufficiently expressive feature space is used.

5.2 Main Findings of the Study

The experimental results indicate that the neural architecture effectively absorbs the fairness penalty by utilizing its high-dimensional latent space $\phi(x; w)$. While the 48-dimensional raw linear space lacks the expressiveness required to satisfy exposure targets without displacing high-reward arms, the 64-dimensional latent representations provide sufficient capacity to accommodate the fairness bonus. A pivotal finding is the increase in the female hit rate ($0.717 \rightarrow 0.726$) under Fair-NeuralLinUCB. This confirms that the model achieves fairness through a “lifting” mechanism, enhancing the visibility of underserved groups, rather than degrading the majority. This is not only a theoretical preference in algorithmic fairness literature but also a significant practical advantage for long-term user retention.

The comparative stability of the models highlights the architectural advantage of the neural approach. FairLinUCB’s wide standard deviation ($\pm 1,031$ in reward compared to ± 477 for Fair-NeuralLinUCB) reflects the inherent instability of forcing rigid linear constraints to balance exposure targets and reward maximization. Conversely, the near-overlapping cumulative regret curves of the neural models demonstrate that the fairness bonus (calibrated at $\beta = 2.0, \epsilon = 0.05$) operates within the effective exploration budget. Within the neural framework, the constraint serves as a targeted adjustment to arm selection rather than a disruptive policy override.

Finally, while FairLinUCB achieves the narrowest absolute gender gap (0.0129), it does so by compressing the distribution, resulting in the lowest hit rates

for both male (0.7126) and female (0.6997) users. This outcome suggests that achieving parity by reducing overall system performance is not desirable from a sociotechnical perspective. The hybrid neural approach, in contrast, offers a more viable path forward for ethical AI by achieving equity through enhanced representation rather than by reducing utility.

5.3 Relation to Previous Research and Originality

This thesis introduces a novel intersection between high-capacity representation learning and fairness-aware decision-making. The significance of these findings is best understood through their relation to the following established paradigms:

- **Extension of the Score-Augmentation Framework:** This research directly extends the work of Huang et al. [24], demonstrating that the additive score-augmentation mechanism employed in Fair-LinUCB can be effectively transferred to a neural representation space. By preserving the UCB score structure via a frozen feature extractor, the study identifies a clear insertion point for fairness signals that is absent in more stochastic frameworks such as Thompson Sampling or ϵ -greedy.
- **Validation of the Deep-Linear Paradigm:** The results validate the effectiveness of the DeepLinUCB architecture proposed by Shi et al. [53]. By demonstrating that the “offline pre-training + frozen weights” design outperforms standalone LinUCB while avoiding the “catastrophic forgetting” risks in other deep algorithms identified by Zahavy and Mannor [62]. This study extends the findings of previous production-scale research (e.g., at Hulu) into ethically constrained environments and provides new evidence for the robustness of neural-linear decoupling.
- **Mitigation of the “Deadly Triad”:** Although recent RL-based fairness systems ([38, 18]) achieve parity, they frequently encounter computational overhead and the “deadly triad” common in full RL. Fair-NeuralLinUCB provides a novel and practically deployable alternative that maintains stability by eliminating the overhead of modeling complex, multi-step state transitions while preserving deep neural feature extraction.

The model achieves comparable group-fairness outcomes while offering superior computational efficiency.

- **Empirical Validation of Dataset Bias:** The unconstrained NeuralLinUCB baseline replicated the demographic recommendation disparities in MovieLens previously documented by Ekstrand et al. [17]. This serves as critical internal validation, confirming that the dataset remains a valid and appropriate testbed for fairness interventions in contemporary latent-factor models.

Original Contribution and Gap Addressed: Prior to this research, the literature was largely divided between non-linear representation learning and active in-processing group-exposure constraints. This thesis addresses this gap by providing the first empirical evidence that integrating these two components within a single pipeline is both computationally feasible and superior to either approach in isolation. By demonstrating that neural feature extraction absorbs the fairness penalty, this work advances the Fairness-Accuracy Pareto Frontier and suggests a new standard for ethical recommendation system design.

5.4 Limitations of the Study

A critical evaluation of the research design identifies several constraints that affect both the interpretation and the broader applicability of the findings.

Validity: The study maintains high internal validity through a controlled simulation environment that uses fixed random seeds, identical candidate sampling across models, and a validated SVD reward oracle. This approach ensures that observed performance differences are attributable to the algorithmic architectures. However, reliance on a deterministic oracle eliminates the stochastic noise present in real-world user feedback, potentially overestimating the models' stability. Furthermore, construct validity is partially constrained by the use of a binary reward threshold (positive reward is ≥ 4). Although this threshold is standard in bandit literature, it discards ordinal preference data and may oversimplify the complexity of user satisfaction.

Reliability and Reproducibility: Reliability is supported by averaging performance across 10 random seeds and conducting significance testing. However, a sample size of 10 seeds is relatively small for normality-based tests (the Shapiro-Wilk test was used as a check). The frequent use of the Wilcoxon signed-rank test as a fallback suggests that the findings are somewhat sensitive to outlier seeds. In terms of reproducibility, although the pipeline and hyperparameters are fully documented, the NVIDIA RTX 5070 GPU environment is not publicly accessible, and floating-point non-determinism across hardware may introduce minor result variance when replicating the study on different computing architectures.

Generalizability and Transferability: The generalizability of these conclusions is limited by the use of the MovieLens 1M dataset. The filtered subset exhibits a notably skewed male-to-female ratio (2.73 : 1), which may limit the applicability of the findings to datasets with balanced demographics or different interaction densities. Although the “frozen neural extractor” design is domain-agnostic, its transferability to high-stakes environments such as healthcare or recruitment remains untested and would require developing domain-specific fairness signals.

The Individual vs. Group Fairness Paradox: A fundamental limitation arises from the conflict between group and individual fairness. Enforcing demographic parity at the group level may paradoxically violate individual fairness. For instance, a female user with a genuine preference for action films may receive fewer such recommendations because the fairness tracker intervenes to reduce the aggregate genre-gender gap. This “individual fairness tax” is an acknowledged and unresolved limitation of all group-parity literature.

5.5 Ethical and Societal Implications

The findings of this study extend beyond technical performance and address the fundamental responsibilities associated with deploying recommendation systems within a diverse society.

The primary ethical contribution of this research is the demonstration that the “fairness tax” is not an inherent constraint but rather a function of model expressiveness. By establishing that a more expressive architecture,

such as the neural-linear model, can minimize utility loss, this study re-frames the policy conversation: the ethical burden lies in selecting an architecture that supports equity without undermining the system’s primary objectives. The adoption of an in-processing design constitutes an ethical intervention. Penalizing bias during arm selection, rather than relying on post-hoc adjustments, prevents biased patterns from entering the feedback loop. This proactive strategy addresses the “rich-get-richer” Matthew Effect, in which popular or majority-favored items gain disproportionate visibility, and then such patterns can compound over time.

The 2.73 : 1 male-to-female ratio in the filtered dataset serves as a cautionary example for practitioners. This finding illustrates how standard activity-based filtering pipelines can inadvertently amplify pre-existing societal imbalances, potentially resulting in a “feedback loop of exclusion” if left unchecked. Although movie recommendations represent a low-stakes application, this architecture is designed for potential transfer to high-stakes contexts, such as medical advice and news distribution. Responsible deployment in such domains requires not only technical efficacy but also regulatory review and stakeholder consultation. Additionally, algorithmic accountability remains constrained by limited explainability, as the fairness parameters (ϵ, β) are technical abstractions that do not yet provide natural-language explanations to non-technical stakeholders of the rationale behind specific recommendation modifications.

5.6 Use of IT and AI Tools in the Thesis

The technical execution and authorship of this thesis were supported by a suite of computational and generative tools, each categorized according to its specific role in the research process.

The experimental pipeline was implemented using Python 3.10. The core library dependencies included PyTorch, NumPy, Pandas, Scikit-learn, and Surprise Library. All simulations were executed in Jupyter Notebooks via VS Code, leveraging NVIDIA RTX 5070 GPU acceleration to handle the computational demands of the neural-linear training cycles. Data visualizations, including cumulative regret curves with 95% confidence intervals and hit rate distributions, were generated using Matplotlib and Seaborn.

Generative AI tools (Google Gemini and Anthropic Claude) were employed as collaborative assistants. Their use was strictly limited to brainstorming support, assisting with organizing existing research, providing clarity on theories, refining prose, and debugging code when needed. It is explicitly stated that all core research components—including the design of the Fair-NeuralLinUCB algorithm, mathematical derivations, the architecture of the simulation environment, the execution of experiments, and the final interpretation of statistical results—are the original work of the author. AI was not used to generate, select, or analyze experimental data. This usage aligns with the university’s academic integrity policies, viewing AI as an assistive tool.

5.7 Future Research

While Fair-NeuralLinUCB successfully demonstrates the benefit of neural representations in constrained environments, multiple directions for future research could further enhance the algorithm’s utility and theoretical basis.

Architectural and Methodological Extensions: A primary technical objective is to derive formal regret bounds. Although empirical results confirm sublinear convergence, establishing theoretical guarantees under the frozen feature extractor assumption, similar to those proven for Fair-LinUCB [24], would strengthen the model’s position within the contextual bandit literature. Methodologically, future research should progress from offline simulations with fixed oracles to live A/B deployments. Such deployments would provide the ecological validity needed to observe non-stationary user behavior and potential feedback loops that arise when users respond to algorithmic fairness interventions in real time.

Evolving the Fairness Framework: The current model can be extended along three dimensions of equity. **Intersectional Fairness:** Future iterations should evolve the Fairness Tracker to monitor intersectional gaps (e.g., gender $\times$ age $\times$ occupation). **Individual Fairness:** Incorporate counterfactual fairness criteria to ensure that individual recommendations remain stable when protected attributes change. **Multi-Sided Fairness:** This

study focused exclusively on the consumer (user-side). A more comprehensive framework would simultaneously enforce item-side (provider) fairness, ensuring equitable exposure for content producers throughout the platform. **Adaptive Control:** Instead of relying on static hyperparameters (such as ϵ and β), a dynamic controller, such as a multi-objective Lagrangian, could be implemented to adjust fairness penalties in real time based on observed exposure drift.

Transparency and Societal Impact: To satisfy regulatory accountability standards in high-stakes domains, integrating explainability methods (e.g., SHAP or integrated gradients) is essential. These methods would enable auditors to identify which latent features trigger fairness penalties for specific arm selections. Additionally, the framework should be evaluated in domains where demographic bias results in material harm, such as clinical trial matching or political news content, to determine domain-specific calibration requirements.

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Appendix A

Code and supplementary materials for this work can be found at: <https://github.com/Pabasara711/FairNeuralLinUCB-Model.git>.